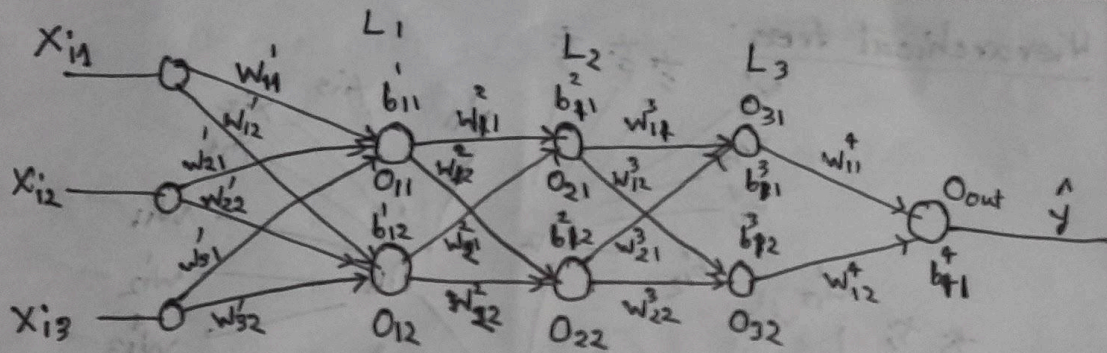


Design a fully connected neural network with 3 input neurons, 3 hidden layers, and 1 output neuron. You may choose the number of neurons in each hidden layer and specify the activation functions used.

1. Define the architecture clearly, including the dimensions of all weight matrices and bias vectors.
2. Initialize all weights and biases with small numerical values.
3. Perform the complete forward pass for one training example and calculate the output step by step.
4. Define a suitable loss function (for example, Mean Squared Error).
5. Derive all gradients manually using backpropagation.
6. Compute the gradients for each weight and bias in every layer.
7. Show the complete backward pass, explaining how the error flows from the output layer back to the first hidden layer.
8. Perform one full parameter update using gradient descent and show the updated weights.

Backpropagation



For L_1 :

$$O_{11} = w_{11}^1 X_{i1} + w_{21}^1 X_{i2} + w_{31}^1 X_{i3} + b_{11}^1$$

$$O_{12} = w_{12}^1 X_{i1} + w_{22}^1 X_{i2} + w_{32}^1 X_{i3} + b_{12}^1$$

For L_2 :

$$O_{21} = w_{11}^2 O_{11} + w_{22}^2 O_{12} + b_{21}^2$$

$$O_{22} = w_{12}^2 O_{11} + w_{22}^2 O_{12} + b_{22}^2$$

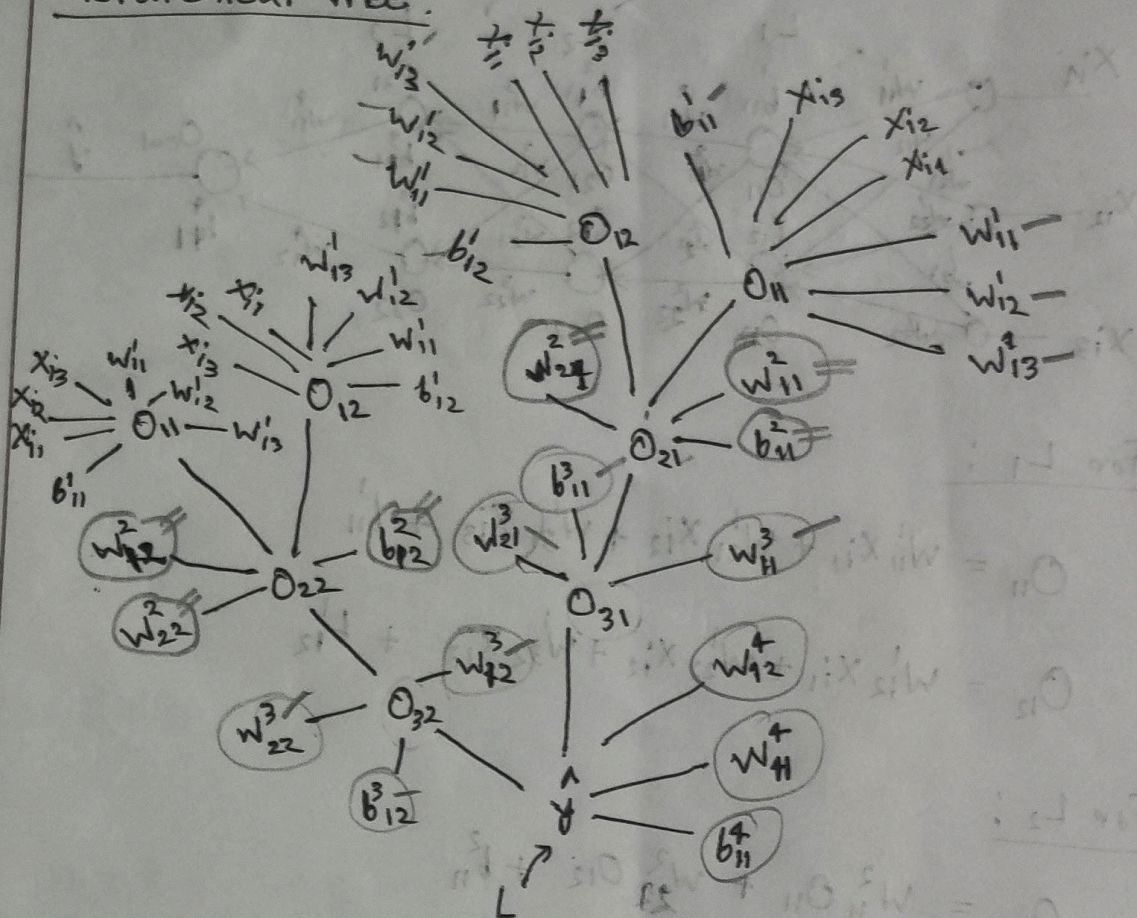
For L_3 :

$$O_{31} = w_{11}^3 O_{21} + w_{21}^3 O_{22} + b_{31}^3$$

$$O_{32} = w_{12}^3 O_{21} + w_{22}^3 O_{22} + b_{32}^3$$

$$\hat{y} = O_{out} = w_{11}^4 O_{31} + w_{12}^4 O_{32} + b_{41}^4$$

Hierarchical tree:



Take (MSE) mean squared error as a Loss function:

$$L = (y - \hat{y})^2 \quad \text{Here } y = \text{Fixed output}$$

$$\hat{y} = \text{Predicted output}$$

Goal is to minimize L .

To minimize loss we have only 2 option that update weights and bias.

From Gradient Descent weight update rule:

$$W_{\text{new}} = W_{\text{old}} - \eta \frac{\partial L}{\partial W_{\text{old}}}$$

$$b_{\text{new}} = b_{\text{old}} - \eta \frac{\partial L}{\partial b_{\text{old}}}$$

we need to find all the weights and bias derivatives now:

For output layer findings:

$$\frac{\partial L}{\partial W_{11}^+}, \frac{\partial L}{\partial W_{12}^+}, \frac{\partial L}{\partial b_{11}^+}$$

$$\frac{\partial L}{\partial W_{11}^+} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial W_{11}^+} \quad [\text{applying chain rule}]$$

$$= \frac{\partial}{\partial \hat{y}} \left[\frac{L}{2} \right] \times \frac{\partial \hat{y}}{\partial W_{11}^+} \quad \begin{matrix} \nearrow \hat{y} \rightarrow W_{11}^+ \\ \nearrow \hat{y} \end{matrix}$$
$$= \frac{\partial}{\partial \hat{y}} \left[\frac{1}{2} (\gamma - \hat{y})^2 \right] \times \frac{\partial}{\partial W_{11}^+} \left[(W_{11}^+ O_{31} + W_{12}^+ O_{32} + b_{11}^+) \right]$$
$$= -2 (\gamma - \hat{y}) O_{31}$$

$$\frac{\partial L}{\partial W_{12}^+} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial W_{12}^+}$$

$$= \frac{\partial}{\partial \hat{y}} \left[\frac{1}{2} (\gamma - \hat{y})^2 \right] \times \frac{\partial}{\partial W_{12}^+} \left[W_{11}^+ O_{31} + W_{12}^+ O_{32} + b_{11}^+ \right]$$
$$= -2 (\gamma - \hat{y}) O_{32}$$

$$\begin{aligned}
 \frac{\partial L}{\partial b_{11}^4} &= \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial b_{11}^4} \\
 &= \frac{\partial}{\partial \hat{y}} [(\gamma - \hat{y})^2] \cdot \frac{\partial}{\partial b_{11}^4} [(w_{11}^4 o_{31} + w_{12}^4 o_{32} + b_{11}^4)] \\
 &= -2(\gamma - \hat{y}) \cdot 1
 \end{aligned}$$

For layer 3:

$$\frac{\partial L}{\partial w_{11}^3}, \frac{\partial L}{\partial w_{21}^3}, \frac{\partial L}{\partial b_{11}^3}, \frac{\partial L}{\partial w_{12}^3}, \frac{\partial L}{\partial w_{22}^3}, \frac{\partial L}{\partial b_{12}^3}$$

$$\frac{\partial L}{\partial w_{11}^3} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial o_{31}} \cdot \frac{\partial o_{31}}{\partial w_{11}^3}$$

$L \rightarrow \hat{y} \rightarrow o_{31} \rightarrow w_{11}^3$

$$\frac{\partial L}{\partial w_{21}^3} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial o_{31}} \cdot \frac{\partial o_{31}}{\partial w_{21}^3}$$

$$\frac{\partial L}{\partial b_{11}^3} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial o_{31}} \cdot \frac{\partial o_{31}}{\partial b_{11}^3}$$

$$\frac{\partial L}{\partial w_{12}^3} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial o_{32}} \cdot \frac{\partial o_{32}}{\partial w_{12}^3}$$

$L \rightarrow \hat{y} \rightarrow o_{32} \rightarrow w_{12}^3$

$$\frac{\partial L}{\partial w_{22}^3} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial o_{32}} \cdot \frac{\partial o_{32}}{\partial w_{22}^3}$$

$$\frac{\partial L}{\partial b_{12}^3} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial o_{32}} \cdot \frac{\partial o_{32}}{\partial b_{12}^3}$$

$$\therefore \frac{\partial O_{31}}{\partial w_{11}^3} = \frac{\partial}{\partial w_{11}^3} [(w_{11}^3 O_{21} + w_{21}^3 O_{22} + b_{11}^3)] = O_{21}$$

$$\frac{\partial O_{31}}{\partial w_{21}^3} = O_{22}$$

$$\frac{\partial O_{31}}{\partial b_{11}^3} = 1.$$

$$\boxed{\frac{\partial L}{\partial \hat{y}} = -2(y - \hat{y})}$$

$$\frac{\partial \hat{y}}{\partial O_{31}} = \frac{\partial}{\partial O_{31}} [(w_{11}^4 O_{31} + w_{12}^4 O_{32} + b_{11}^4)] = w_{11}^4$$

$$\frac{\partial O_{32}}{\partial w_{12}^3} = \frac{\partial}{\partial w_{12}^3} [(w_{12}^3 O_{21} + w_{22}^3 O_{22} + b_{12}^3)] = O_{21}$$

$$\frac{\partial O_{32}}{\partial w_{22}^3} = O_{22}$$

$$\frac{\partial O_{32}}{\partial b_{12}^3} = 1$$

$$\frac{\partial \hat{y}}{\partial O_{32}} = \frac{\partial}{\partial O_{32}} [(w_{11}^4 O_{31} + w_{12}^4 O_{32} + b_{11}^4)] = w_{12}^4$$

For L3:

$$\frac{\partial L}{\partial w_{11}^3} = -2(y - \hat{y}) \cdot w_{11}^4 \cdot O_{21}$$

$$\frac{\partial L}{\partial w_{12}^3} = -2(y - \hat{y}) \cdot w_{12}^4 \cdot O_{21}$$

$$\frac{\partial L}{\partial w_{21}^3} = -2(y - \hat{y}) \cdot w_{11}^4 \cdot O_{22}$$

$$\frac{\partial L}{\partial w_{22}^3} = -2(y - \hat{y}) \cdot w_{12}^4 \cdot O_{22}$$

$$\frac{\partial L}{\partial b_{11}^3} = -2(y - \hat{y}) \cdot w_{11}^4$$

$$\frac{\partial L}{\partial b_{12}^3} = -2(y - \hat{y}) \cdot w_{12}^4$$

For layer 2:

$$\frac{\partial L}{\partial w_{11}^2}, \frac{\partial L}{\partial w_{21}^2}, \frac{\partial L}{\partial b_{11}^2}, \frac{\partial L}{\partial w_{12}^2}, \frac{\partial L}{\partial w_{22}^2}, \frac{\partial L}{\partial b_{12}^2}$$

$$\frac{\partial L}{\partial w_{11}^2} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_{31}} \frac{\partial o_{31}}{\partial w_{11}^2} \frac{\partial o_{21}}{\partial o_{21}} \frac{\partial o_{21}}{\partial w_{11}^2}$$

$$L \rightarrow \hat{y} \rightarrow o_{31} \rightarrow o_{21} \rightarrow w_{11}^2$$

$$\frac{\partial L}{\partial w_{21}^2} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_{31}} \frac{\partial o_{31}}{\partial o_{21}} \frac{\partial o_{21}}{\partial w_{21}^2}$$

$$\frac{\partial L}{\partial b_{11}^2} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_{31}} \frac{\partial o_{31}}{\partial o_{21}} \frac{\partial o_{21}}{\partial b_{11}^2}$$

$$\frac{\partial L}{\partial w_{12}^2} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_{32}} \frac{\partial o_{32}}{\partial o_{22}} \frac{\partial o_{22}}{\partial w_{12}^2}$$

$$L \rightarrow \hat{y} \rightarrow o_{32} \rightarrow o_{22} \rightarrow w_{12}^2$$

$$\frac{\partial L}{\partial w_{22}^2} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_{32}} \frac{\partial o_{32}}{\partial o_{22}} \frac{\partial o_{22}}{\partial w_{22}^2}$$

$$\frac{\partial L}{\partial b_{12}^2} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_{32}} \frac{\partial o_{32}}{\partial o_{22}} \frac{\partial o_{22}}{\partial b_{12}^2}$$

Here,

$$\frac{\partial \hat{y}}{\partial o_{31}} = \frac{\partial}{\partial o_{31}} [(w_{11}^4 o_{31} + w_{12}^4 o_{32} + b_{11}^4)] = w_{11}^4$$

$$\frac{\partial o_{31}}{\partial o_{21}} = \frac{\partial}{\partial o_{21}} [(w_{11}^3 o_{21} + w_{21}^3 o_{22} + b_{11}^3)] = w_{11}^3$$

$$\frac{\partial o_{21}}{\partial w_{11}^2} = \frac{\partial}{\partial w_{11}^2} [(w_{11}^2 o_{11} + w_{22}^2 o_{12} + b_{11}^2)] = o_{11}$$

$$\frac{\partial \hat{y}}{\partial o_{32}} = \frac{\partial}{\partial o_{32}} [(w_{11}^4 o_{31} + w_{12}^4 o_{32} + b_{11}^4)] = w_{12}^4$$

$$\frac{\partial o_{32}}{\partial o_{22}} = \frac{\partial}{\partial o_{22}} [(w_{12}^3 o_{21} + w_{22}^3 o_{22} + b_{12}^3)] = w_{22}^3$$

$$\frac{\partial o_{22}}{\partial w_{12}^2} = \frac{\partial}{\partial w_{12}^2} [(w_{12}^2 o_{21} + w_{22}^2 o_{12} + b_{12}^2)] = o_{11}$$

For L2:

$$\frac{\partial L}{\partial w_{11}^2} = -2(\hat{y} - y) w_{11}^4 w_{11}^3 o_{11}$$

$$\frac{\partial L}{\partial w_{12}^2} = -2(\hat{y} - y) w_{12}^4 w_{22}^3 o_{11}$$

$$\frac{\partial L}{\partial w_{21}^2} = -2(\hat{y} - y^4) w_{11}^4 w_{11}^3 o_{22}$$

$$\frac{\partial L}{\partial w_{22}^2} = -2(\hat{y} - y) w_{12}^4 w_{22}^3 o_{12}$$

$$\frac{\partial L}{\partial b_{11}^2} = -2(\hat{y} - y^4) w_{11}^4 w_{11}^3$$

$$\frac{\partial L}{\partial b_{12}^2} = -2(\hat{y} - y) w_{12}^4 w_{22}^3$$

For layer 1:

$$\frac{\partial L}{\partial w_{11}^1}, \frac{\partial L}{\partial w_{12}^1}, \frac{\partial L}{\partial w_{13}^1}, \frac{\partial L}{\partial b_{11}^1}, \frac{\partial L}{\partial w_{11}^1}, \frac{\partial L}{\partial w_{12}^1}, \frac{\partial L}{\partial w_{13}^1}, \frac{\partial L}{\partial b_{12}^1}$$

$$\frac{\partial L}{\partial w_{11}^1} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_{31}} \frac{\partial o_{31}}{\partial o_{21}} \frac{\partial o_{21}}{\partial o_{11}} \frac{\partial o_{11}}{\partial w_{11}^1}$$

$$\Rightarrow \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_{31}} \frac{\partial o_{31}}{\partial o_{21}} \frac{\partial o_{21}}{\partial o_{12}} \frac{\partial o_{12}}{\partial w_{11}^1}$$

$$\Rightarrow \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_{32}} \frac{\partial o_{32}}{\partial o_{22}} \frac{\partial o_{22}}{\partial o_{12}} \frac{\partial o_{12}}{\partial w_{12}^1}$$

$$\Rightarrow \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_{32}} \frac{\partial o_{32}}{\partial o_{22}} \frac{\partial o_{22}}{\partial o_{11}} \frac{\partial o_{11}}{\partial w_{11}^1}$$

$$\begin{aligned} L &\rightarrow \hat{y} \rightarrow o_{31} \rightarrow o_{21} \rightarrow o_{11} \rightarrow w_{11}^1 \\ L &\rightarrow \hat{y} \rightarrow o_{31} \rightarrow o_{21} \rightarrow o_{12} \rightarrow w_{11}^1 \\ L &\rightarrow \hat{y} \rightarrow o_{32} \rightarrow o_{22} \rightarrow o_{12} \rightarrow w_{11}^1 \\ L &\rightarrow \hat{y} \rightarrow o_{32} \rightarrow o_{22} \rightarrow o_{11} \rightarrow w_{11}^1 \end{aligned}$$

$$\frac{\partial L}{\partial w'_{12}} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_{31}} \frac{\partial o_{31}}{\partial o_{21}} \frac{\partial o_{21}}{\partial o_{11}} \frac{\partial o_{11}}{\partial w'_{12}}$$

$$L \rightarrow \hat{y} \rightarrow o_{31} \rightarrow o_{21} \rightarrow o_{11} \rightarrow w'_{12}$$

$$\Rightarrow \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_{31}} \frac{\partial o_{31}}{\partial o_{21}} \frac{\partial o_{21}}{\partial o_{12}} \frac{\partial o_{12}}{\partial w'_{12}}$$

$$L \rightarrow \hat{y} \rightarrow o_{31} \rightarrow o_{21} \rightarrow o_{12} \rightarrow w'_{12}$$

$$L \rightarrow \hat{y} \rightarrow o_{32} \rightarrow o_{22} \rightarrow o_{11} \rightarrow w'_{12}$$

$$\Rightarrow \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_{32}} \frac{\partial o_{32}}{\partial o_{22}} \frac{\partial o_{22}}{\partial o_{11}} \frac{\partial o_{11}}{\partial w'_{12}}$$

$$L \rightarrow \hat{y} \rightarrow o_{32} \rightarrow o_{22} \rightarrow o_{12} \rightarrow w'_{12}$$

$$\Rightarrow \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_{32}} \frac{\partial o_{32}}{\partial o_{22}} \frac{\partial o_{22}}{\partial o_{12}} \frac{\partial o_{12}}{\partial w'_{12}}$$

$$\frac{\partial L}{\partial w'_{13}} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_{31}} \frac{\partial o_{31}}{\partial o_{21}} \frac{\partial o_{21}}{\partial o_{11}} \frac{\partial o_{11}}{\partial w'_{13}}$$

$$L \rightarrow \hat{y} \rightarrow o_{31} \rightarrow o_{21} \rightarrow o_{11} \rightarrow w'_{13}$$

$$L \rightarrow \hat{y} \rightarrow o_{31} \rightarrow o_{21} \rightarrow o_{12} \rightarrow w'_{13}$$

$$L \rightarrow \hat{y} \rightarrow o_{32} \rightarrow o_{22} \rightarrow o_{11} \rightarrow w'_{13}$$

$$\Rightarrow \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_{31}} \frac{\partial o_{31}}{\partial o_{21}} \frac{\partial o_{21}}{\partial o_{12}} \frac{\partial o_{12}}{\partial w'_{13}}$$

$$L \rightarrow \hat{y} \rightarrow o_{32} \rightarrow o_{22} \rightarrow o_{12} \rightarrow w'_{13}$$

$$\Rightarrow \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_{32}} \frac{\partial o_{32}}{\partial o_{22}} \frac{\partial o_{22}}{\partial o_{11}} \frac{\partial o_{11}}{\partial w'_{13}}$$

$$\Rightarrow \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_{32}} \frac{\partial o_{32}}{\partial o_{22}} \frac{\partial o_{22}}{\partial o_{12}} \frac{\partial o_{12}}{\partial w'_{13}}$$

$$\frac{\partial L}{\partial b'_{11}} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_{31}} \frac{\partial o_{31}}{\partial o_{21}} \frac{\partial o_{21}}{\partial o_{11}} \frac{\partial o_{11}}{\partial b'_{11}}$$

$$L \rightarrow \hat{y} \rightarrow o_{31} \rightarrow o_{21} \rightarrow o_{11} \rightarrow b'_{11}$$

$$L \rightarrow \hat{y} \rightarrow o_{32} \rightarrow o_{22} \rightarrow o_{11} \rightarrow b'_{11}$$

$$L \rightarrow \hat{y} \rightarrow o_{31} \rightarrow o_{21} \rightarrow o_{12} \rightarrow b'_{12}$$

$$\Rightarrow \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_{32}} \frac{\partial o_{32}}{\partial o_{22}} \frac{\partial o_{22}}{\partial o_{11}} \frac{\partial o_{11}}{\partial b'_{11}}$$

$$L \rightarrow \hat{y} \rightarrow o_{32} \rightarrow o_{22} \rightarrow o_{12} \rightarrow b'_{12}$$

$$\frac{\partial L}{\partial b'_{12}} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_{31}} \frac{\partial o_{31}}{\partial o_{21}} \frac{\partial o_{21}}{\partial o_{12}} \frac{\partial o_{12}}{\partial b'_{12}}$$

$$\Rightarrow \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_{32}} \frac{\partial o_{32}}{\partial o_{22}} \frac{\partial o_{22}}{\partial o_{12}} \frac{\partial o_{12}}{\partial b'_{12}}$$

Rest works left , need to assign the values and calculations are left like using sigmoid, calculate weighted sum and forward pass y pred , then calculating the loss, after getting the loss (using mse) value, need to update the new weights and bias using the gradient descent where i already calculated each gradients.